



OFFICIAL

NMHS MHS GUIDELINE

Medication: Prescribing and Administration of Antipsychotic Long-Acting Injections

Scope (Audience)	All NMHS Mental Health Services (MHS) Clinical Staff
Scope (Area)	All NMHS MHS Areas
Summary	Provides NMHS Mental Health Services (MHS) clinical staff with guidelines for prescribing and administration of long-acting injection (LAI) antipsychotics to ensure safe and effective treatment for patients of the service.
Risk Rating	Medium
NSQHS Standards	Clinical Governance (Std 1) Medication Safety (Std 4) Comprehensive Care (Std 5)
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Policy Contact	Chief Pharmacist MHSDHS
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1. Aim

This document provides NMHS Mental Health Services (MHS) clinical staff with guidelines for prescribing and administration of long-acting injection (LAI) antipsychotics to ensure safe and effective treatment for patients of the service.

2. Background

Intramuscular (IM) long-acting, or depot antipsychotics are medication formulations designed to release medication gradually over a period of time. Benefits may include more stable medication plasma levels, convenience and a reduced risk of relapse due to non-adherence.

The decision to initiate a LAI antipsychotic should be based on the patient's individual circumstances and a clinical assessment of the risks and consequences of medication non-adherence.

3. Risk

Patients may suffer adverse outcomes related to LAIs if these guidelines are not followed. Non-compliance with this procedure will impact on the following risk categories:

Culture	☐	Organisational Development	☐
Emergency Management	☐	Policy & Legislative Compliance	☐
Fiscal Responsibility	☐	Quality of Patient Care	☒
Governance & Accountability	☐	Research & Innovation	☐
Information Communications Technology	☐	Staff Safety & Wellness	☐
Facility Management	☐	Workforce Planning	☐
Medical Equipment	☐	Other	☐

4. Procedure

4.1 Initiation of Depot/ LAI

- When initiating long-acting therapy, tolerability must be tested at a minimum prior to administration of the injection if they have never had prior exposure to the medication via any route.
- Oral medication must be trialled in order to rule out any potential anaphylactic or allergic reaction to the medication
- Preferably oral medication should be trialled to test medication efficacy, but it is acknowledged in some circumstances this may not be possible
- LAI preparations in an oily base (e.g. haloperidol, zuclopenthixol, flupentixol (flupenthixol)) require a test dose of medication given 4-10 days before commencing a maintenance dose. This is to also test tolerability to the oily base as well as the medication. This test dose may be administered in place of an oral test dose if there is no oral formulation available (e.g. flupentixol (flupenthixol) decanoate).
- Some LAI also have specific requirements for loading regimes or concomitant oral supplementation. These are documented under the individual medication monograph

- Consideration must be given to maximum licensed dosing and antipsychotic polypharmacy and documented as per the [NMHS MH Medication: Prescribing Greater than the Maximum Dose](#).

4.2 General Intramuscular Administration

- Gently aspirate the syringe pre-injection to ensure the needle has not entered a blood vessel. If this occurs withdraw the needle and repeat the process with a sterile needle.
- Z-track technique should be used, particularly with typical antipsychotic LAIs.
- Record the site and side of administration on the Intramuscular Long-Acting Injection Chart - PMR58.
- Rotate the site of injection to prevent damage, protect the administration route and maximise patient comfort.
- In general, a maximum volume of 3 mL may be administered into the gluteal muscle and a maximum of 2 mL may be administered into the deltoid muscle. Some licensed pre-filled LAI syringes may have volumes that are over this threshold but may be administered safely as per manufacturer's instructions (eg. paliperidone palmitate Trinza® and Hafyera® formulations).

4.3 Site of Injection

All LAIs are to be administered by deep intramuscular injection. Most LAIs are licensed for administration in specific site(s). Administration in any other site is deemed off-label and rationale must be documented in the patient's integrated progress notes.

4.3.1 Gluteal injection site

Ventrogluteal injection site: The ventrogluteal injection site is located midway between the two bone structures of the hip and head of femur. The ventrogluteal site is the injection site of choice for gluteal intramuscular antipsychotic injections as:

- There are bony landmarks that are considered easy to palpate
- Greater thickness of muscle than the dorsogluteal site
- It has a thinner layer of subcutaneous fat, making inadvertent subcutaneous injection less likely
- The site is relatively free of nerves and blood vessels, reducing the potential for more significant injury.

Dorsogluteal injection site: The dorsogluteal site is located in the upper outer quadrant of the buttock, in the superior lateral aspect of the gluteal muscles. This site has historically been the most commonly used, but it is **not** the preferred site of gluteal injection due to:

- The presence of major nerves (including the sciatic nerve) and blood vessels in this area
- The possibility of a thick layer of adipose tissue, increasing the risk of subcutaneous injection.

4.3.2 Deltoid injection site

A deltoid injection should be given at, or immediately below, the mid-point of the deltoid muscle, and should never be given below the level of the axilla (armpit).

Pain appears to be one of the most frequently reported complications associated with this site. The deltoid has the greatest blood flow of any muscle routinely used for IM injections meaning it is preferred for rapid onset injections.

4.4 Missed doses

- If a LAI dose is missed, it should be given as soon as possible after the due date.
- The maximum acceptable delay and the need for dose adjustment or 'top up' oral dosing is related to the dose and frequency at which the LAI is prescribed.

4.4.1 Missed doses of LAI antipsychotics are managed in a stepwise manner

Step 1 – Determine how many injections the patient received prior to the last dose (initiation and maintenance phase)

Step 2 – Ascertain the date of the last injection

Step 3 – Calculate the time that has passed since the last injection

Step 4 – Manage missed injection by:

- Administering an immediate dose of the injection as soon as possible
- Supplementing the injection with oral antipsychotic, or
- Re-establishing oral antipsychotic therapy and then initiating a LAI.

Specific guidance for missed doses is given under each medication. Further information can be obtained from your Clinical Pharmacist or the Medicines Information Service.



5. Individual Medication Guidelines

5.1 Risperidone

For dosing equivalence between risperidone oral, LAI and paliperidone – see table below: **Approximate dose equivalence between risperidone and paliperidone formulations**

Medication	Available products	Licensed injection site	Initiation regime	Licensed maintenance dose range	Accepted exceptional dosing window	Storage
Risperidone microspheres	Risperdal Consta® 25mg, 37.5mg, 50mg	Deltoid or gluteal	Oral antipsychotic supplementation required for 21 days from 1st injection (or any dose increase)	25mg - 50mg every 2 weeks	Nil. Dosing needs to be maintained at 2-week intervals Release is delayed by 3 weeks, maintained from weeks 4-6 and subsides by week 7.	Stored in fridge (2-8°C). Bring to room temperature 30 mins before reconstituting and administering. May be kept out of fridge for 7 days (unreconstituted) and for 6hrs after reconstitution (Discard after this time). Do not re-refrigerate reconstituted product.

Risperdone modified release	Risvan® 75mg 100mg	Deltoid or gluteal	No initiation regimen required	75mg-100mg every 28 days	May be given up to 3 days before next due date.	Store at room temperature (25°C)
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Preparation and Administration

Risperdal Consta®:

- Risperdal Consta® vials must be reconstituted using the supplied diluent and needles.
 - Deltoid Injection: Use the 1 inch/25mm 21G needle
 - Gluteal injection: Use the 2 inches/50mm, 20G needle
- The entire contents of the vial must be administered (partial vial doses are inaccurate and must not be prescribed)
 - The maximum licensed dose of Risperidone Consta® is 50mg fortnightly. Doses greater than this are off-label. Doses of 62.5mg or 75mg may be prepared by reconstituting the contents of the relevant two vials with one vial of 2mL diluent. The contents of both vials are withdrawn and administered as a single injection.
 - For a 62.5mg dose, use 25mg & 37.5mg strength vials.
 - For a 75mg dose, either 2 x 37.5mg strengths, or 50mg & 25mg can be used.
 - Doses higher than 75mg must be reconstituted as separate vials and administered as two injections.

Risvan®: **SEE APPENDIX 1 FOR FULL RISVAN® RECONSTITUTION INSTRUCTIONS**

- Risvan® must be reconstituted using the supplied components and diluent in the kit box
 - Deltoid injection: Use the 1 inch/25mm 21G needle (green cap)
 - Gluteal injection: Use the 2 inches/50mm, 20G needle (yellow cap)
- Check that the diluent is a free-flowing clear liquid and not viscous or gel-like. If not, rub diluent syringe between the hands to warm it up until it is a free-flowing liquid.

- The entire content in the syringe must be administered (partial doses are inaccurate and must not be prescribed)
 - The maximum licensed dose of Risvan® is 100mg every 28 days. Doses greater than this are off-label. Doses higher than 100mg will need to be administered as two separate injections into different sites.

Missed Doses

Risperdal Consta® Missed Doses			
Time since Last Injection	At steady state (5 injections) and ≤6 weeks since last injection	When steady state not reached (<5 injections) and >2 weeks since last dose	When steady state (≥5 injections) has been reached and >6 weeks since last injection
Action	• Give next injection as soon as possible	• Give next injection as soon as possible plus oral supplementation for 3 weeks	• Give next injection as soon as possible plus oral supplementation for 3 weeks
Risvan® Missed Doses			
Action	Give next injection as soon as possible. No oral supplementation required (Risvan® has a rapid release of risperidone).		

5.2 Paliperidone Palmitate (Invega Sustenna®, Invega Trinza® and Invega Hafyera®)

Paliperidone palmitate is available in three formulations – monthly, 3 monthly and 6 monthly preparations. Care must be taken to prescribe and select the correct preparation. To distinguish between the formulations, the BRAND name MUST be annotated on medication orders and prescriptions (eg. Sustenna®, Trinza® or Hafyera®).

Medication	Available products	Licensed injection site	Initiation regime	Licensed maintenance dose range	Accepted exceptional dosing window	Storage
Paliperidone Palmitate	Invega Sustenna® 25mg, 50mg, 75mg, 100mg, 150mg	Deltoid or gluteal	Loading regime: 150mg on Day 1 and 100mg on Day 8 then maintenance from Day 36	25mg -150mg monthly (4 weekly) (Dose adjustment required in mild renal impairment)	May be given +/- 4 days for initiation dose 2 +/- 7 days from next due date	Store at room temperature (25°C)
	Invega Trinza® 175mg, 263mg, 350mg, 525mg	Deltoid or Gluteal	Must be stabilised on 4 doses of Sustenna® (with the last 2 doses of the same strength strongly recommended)	175mg – 525mg 3 monthly (12 weekly) (Dose adjustment required in mild renal impairment)	May be given +/- 14 days from next due date	Store at room temperature (25°C)
	Invega* Hafyera® 700mg, 1000mg	Gluteal (Due to LARGE volume to inject)	Must be stabilised on 4 doses of Sustenna® (with the last 2 doses of the same strength strongly recommended) OR 1 dose of Trinza®	700mg – 1000mg 6 monthly (24 weekly) (Dose adjustment required in mild renal impairment)	May be given +21 days / -14 days from next due date	Store at room temperature (25°C) Product must be stored horizontally to aid reconstitution

Preparation and Administration

Paliperidone palmitate formulations are based on a microcrystalline technology. Differing crystal sizes account for the release properties over time. Each formulation has specific preparation requirements that **must be followed** in order to prevent incomplete administration into the muscle (and hence increase the risk of relapse).

Formulation	Preparation	Administration – ONLY USE THE NEEDLES SUPPLIED. FAILURE TO DO SO WILL RESULT IN NEEDLE BLOCKAGE	
Invega Sustenna®	Prefilled syringe With the tip pointing up, Shake for 10 seconds prior to injecting Ensure homogenous suspension	Deltoid	< 90kg: 1 inch (25mm) 23G needle
			≥ 90kg: 1½ inch (38mm) 22G needle
Invega Trinza®	Prefilled syringe With the tip pointing up, shake VIGOROUSLY for at least 15 seconds within 5 mins prior to injecting Ensure homogenous suspension	Gluteal: 1½ inch (38mm) 22G needle	
		Deltoid	< 90kg: 1 inch (25mm) 22G needle ≥ 90 kg: 1½ inch (38mm) 22G needle
Invega Hafyera®	Prefilled syringe With the tip pointing up, shake VERY FAST for at least 15 seconds, rest briefly then shake again for a further 15 seconds. Ensure homogenous suspension	Gluteal only: 1½ inch (38mm) 20G needle* Note: the large volumes (3.5mL and 5mL) – may cause injection site pain	

* To reduce the risk of needle blockage, do not use needles from the 1-month or 3-month paliperidone palmitate injectable product packs or other commercially available needles.

Missed Doses

Invega Sustenna® Missed Doses						
Time since Last Injection	Initiation phase			Maintenance phase		
	<4 weeks since first injection	Between 4 to 7 weeks since the first injection	>7 weeks since the first injection	4-6 weeks since the last injection	> 6 weeks to 6 months since the last injection	>6 months since the last injection
Action	Give second loading dose (100mg) as soon as possible (deltoid). Then administer 75mg 5 weeks after the first injection. Commence maintenance monthly doses thereafter	Give 2 doses of 100mg in the deltoid 1 week apart. Give maintenance injection 28 days later after the second loading dose (100mg) in either the gluteal or deltoid muscles	Restart the initiation phase (150mg on day 1 then 100mg on day 8)	Give the next maintenance injection as soon as possible then resume usual 4 weekly dose	Give the maintenance injection (deltoid) as soon as possible & then again 1 week later (deltoid) [except for a maintenance dose of 150mg – the patient should receive 2 doses of 100mg (day 1 then day 8 - deltoid) & then 150mg 28 days later]	The patient should undergo re-initiation
Invega Trinza® Missed Doses						

Time since Last Injection	<4 months since the last injection	>4 months up to 9 months:				>9 months since last injection
Action	Give the maintenance dose as soon as possible	- Give 2 booster doses of Invega Sustenna®, 7 days apart (deltoid), as per the following table - Resume the maintenance Invega Trinza® dose 28 days later (deltoid or gluteal)				- Restart on Invega Sustenna® - Invega Trinza® may be reconsidered once the patient is stabilised on Invega Sustenna® for at least 4 months
		Last Invega Trinza® 3-month injection dose	Administer paliperidone palmitate Invega Sustenna®, 2 doses one week apart (deltoid)		Then administer Invega Trinza® 3-month injection dose (deltoid or gluteal)	
			Day 1	Day 8	Day 36 and 12 weekly thereafter	
		175mg	50mg	50mg	175mg	
		263mg	75mg	75mg	263mg	
		350mg	100mg	100mg	350mg	
		525mg	100mg	100mg	525mg	

Invega Hafyera® Missed Doses

Time since Last Injection	>6 months + 3 weeks and < 8 months since last injection			8 – 11 months since last injection			> 11 months since last injection	
Action	• Re-initiate as per table below			• Re-initiate as per table below			Reinitiate with 4 months of Invega Sustenna®	
	Last dose of Invega Hafyera®	Administer Invega Sustenna® 1-monthly into deltoid muscle	Then administer Invega Hafyera® into gluteal muscle	Last dose of Invega Hafyera®	Administer Invega Sustenna® 1-monthly into deltoid muscle			Then administer Invega Hafyera® into gluteal muscle
		Day 1	Day 28 and 6 monthly thereafter		Day 1	Day 8		
	700mg	100mg	700mg	700mg	100mg	100mg		700mg
	1000mg	150mg	1000mg	1000mg	100mg	100mg		1000mg

5.3 Approximate dose equivalence between risperidone and paliperidone formulations

- Paliperidone is the active metabolite of risperidone. The patient should receive a test dose of oral risperidone or oral paliperidone prior to commencing paliperidone LAI if they have not had a trial of either risperidone or paliperidone in any form previously.
- The relationship between the different formulations of paliperidone and doses of risperidone is tabulated below. Please note not all of these preparations can be used interchangeably. Refer to each preparation product information and indication.

Risperidone oral	Paliperidone oral	Risperidone LAI (Risperidal Consta®)	Risperidone LAI (Risvan®)	Paliperidone palmitate LAI 1 monthly (SUSTENNA®)	Paliperidone palmitate LAI 3 monthly (TRINZA®)	Paliperidone palmitate LAI 6 monthly (HAFYERA®)
2 mg/day	4 mg/day (~3mg)	25 mg/2 weeks	N/A	50 mg/4 weeks	175mg/12 weeks	N/A
3 mg/day	6 mg/day	37.5 mg/2 weeks	75mg/4 weeks	75 mg/4 weeks	263mg/12 weeks	N/A
4 mg/day	9 mg/day	50 mg/2 weeks	100mg/4 weeks	100 mg/4 weeks	350mg/12 weeks	700mg/24 weeks
6 mg/day	12 mg/day	-	N/A	150 mg/4 weeks	525mg/12 weeks	1000mg/24 weeks

5.4 Olanzapine pamoate monohydrate

Medication	Available products	Licensed injection site	Initiation regime	Licensed maintenance dose range	Accepted exceptional dosing window	Storage
Olanzapine pamoate monohydrate	Zyprexa Relprevv® 210mg, 300mg, 405mg	Gluteal	Loading Regime for 8 weeks Supplementation with oral olanzapine after the first injection is not necessary	150mg – 300mg every 2 weeks 300mg – 405mg every 4 weeks	No allowance suggested in the product information. 1-2 days before or after the next due dose is acceptable	Store at room temperature (25°C) Once reconstituted, discard after 6 hours.

Converting from oral olanzapine to olanzapine pamoate monohydrate LAI:

Total oral olanzapine dose	Recommended starting dose of olanzapine pamoate (for 8 weeks)	Maintenance dose after 8 weeks of starting olanzapine pamoate treatment
10mg per day	210mg every 2 weeks or 405mg every 4 weeks	150mg every 2 weeks or 300mg every 4 weeks
15mg per day	300mg every 2 weeks	210mg every 2 weeks or 405mg every 4 weeks
20mg per day	300mg every 2 weeks	300mg every 2 weeks

Preparation and Administration

Olanzapine pamoate monohydrate is a crystalline salt formulation that is suspended in water. When injected into muscle, the crystals slowly dissolve and dissociate, releasing olanzapine.

- Olanzapine pamoate vials must be reconstituted using the supplied diluent, syringe and needles.
 - Non-Obese patients: Use the 1½ inch (38mm) 19G needle
 - Obese patients: Use the 2 inch (50mm) 19G needle
 - (The approved product information does not define the terms “non-obese patients” and “obese patients”, so clinician discretion is advised.)
- Once reconstituted as per the table below, the resulting olanzapine concentration is always 150mg/mL
 - There will always be left over suspension in the vial once the appropriate dose has been taken out

Reconstitution Instructions

Dose	Olanzapine pamoate vial strength	Volume of diluent to add (mL)	Final volume to inject (mL)
150mg	210mg	1.3mL	1.0mL
210mg	210mg	1.3mL	1.4mL
300mg	300mg	1.8mL	2.0mL
405mg	405mg	2.3mL	2.7mL

- Prior to administration, aspirate the syringe. If blood is seen, discard the product and needle and reconstitute another solution.
- Administer via deep IM injection into the **gluteal** muscle. This is to minimise the risk of Post Injection Syndrome.

Post Injection Syndrome (PIS) and Monitoring Requirements

- PIS is a rare side effect resulting from inadvertently injecting olanzapine pamoate into the bloodstream
- Onset of this phenomenon is relatively rare (approximately 1.85% of recipients and 0.07% of all injections administered) and typically appears within the first 1 - 2 hours of administration
- Symptoms present similar to an olanzapine overdose and include dizziness, sedation, confusion, delirium and altered gait
- The same risk is present with each injection thus due care must be given to administration and monitoring of this depot for EVERY administration
- After each injection **appropriately trained personnel* must monitor for alertness** every 30 minutes for at least 2 hours**
 - * The approved product information does not define the term “appropriately trained personnel”. Clinician discretion is advised.
 - ** Monitoring involves confirming that the patient is alert, oriented, and absent of any signs and symptoms of overdose and is able to have a small conversation.
 - Measuring the patient’s blood pressure, temperature or pulse does not predict the onset of PIS and is not required.
 - If symptoms of PIS occur, medical supervision and monitoring must be instituted until examination indicates that signs and symptoms have resolved. Referral to an emergency department may be required.
 - For the remainder of the day, patients should be advised to not drive a car nor operate machinery, to be vigilant for signs/symptoms of overdose and be able to access medical treatment if required.
- Patients must agree to the monitoring requirements above. Any patient who does not comply with the monitoring requirements will have their olanzapine LAI reviewed by their treating team and a plan to ensure compliance with monitoring must be established prior to the next planned administration date.

Missed Doses

Zyprexa Relprevv® Missed Doses		
Time since Last Injection	≤8 weeks since the last injection	>8 weeks
Action	Give the next injection as soon as possible. Oral supplementation may be needed if symptoms appear	The patient should receive recommended loading dose for 8 weeks as per the equivalence table above

5.5 Aripiprazole (Abilify Maintena®, Abilify Asimtufii®)

Medication	Available products	Licensed injection site	Initiation regime	Licensed maintenance dose range	Accepted exceptional dosing window	Storage
Aripiprazole monohydrate	Abilify Maintena® 300mg, 400mg	Deltoid or gluteal	Oral antipsychotic supplementation required for 14 days from 1 st injection	400mg every 4 weeks Dose reduction to 300mg or 200mg recommended if adverse reactions, is a known CYP2D6 poor metaboliser or is taking concomitant CYP2D6 and/or CYP3A4 strong inhibitors for more than 14 days	Can be given early from day 26	Store at room temperature (below 25°C) Discard reconstituted vial after 4 hours
Aripiprazole monohydrate	Abilify Asimtufii® 720mg, 960mg	Gluteal only	At least 1 previous dose of Abilify Maintena® 1 monthly LAI, then switch directly to Asimtufii® 2 monthly at next depot due date	960mg every 8 weeks Dose reduction to 720mg recommended if adverse reactions, is a known CYP2D6 poor metaboliser or is taking concomitant CYP2D6 and/or CYP3A4 strong inhibitors for more than 14 days	Can be given up to 2 weeks before, or 2 weeks after the scheduled dose. The following dose should be given 8 weeks (56 days) after the	Store at room temperature (below 25°C)

					administration date	
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5.6 Preparation and Administration

Aripiprazole monohydrate is a lyophilised preparation that is suspended in water for injection

- Aripiprazole monohydrate vials must be reconstituted using the supplied diluent and needles.
- Abilify Maintena®:
 - Obese patients
 - Deltoid: Use the 1.5inch (38mm) 22G needle
 - Gluteal: Use the 2 inch (50mm) 21G needle
 - Non obese patients:
 - Deltoid: Use the 1 inch (25mm), 23G needle
 - Gluteal: Use the 1.5inch (38mm), 22G needle
- Abilify Asimtufii®:
 - Obese patients
 - Gluteal: Use the 2 inch (51mm) 21G needle
 - Non obese patients:
 - Gluteal: Use the 1.5inch (38mm), 22G needle
- Reconstitute Abilify Maintena® as per the table below
 - There will be left over suspension in the vial once the appropriate dose has been taken out.

Reconstitution Instructions for Ability Maintena® 1-monthly LAI only (Asimtufii® is not reconstituted)

Dose to be given	Aripiprazole monohydrate vial strength	Volume of diluent to add (mL)*	Final volume to inject (mL)
400mg	400mg	1.9mL	2.0mL
300mg	400mg	1.9mL	1.5mL
300mg	300mg	1.5mL	1.5 mL
200mg	400mg	1.9mL	1.0mL
200mg	300mg	1.5mL	1.0mL
160mg	400mg	1.9mL	0.8mL
160mg	300mg	1.5mL	0.8mL



Missed Doses

Abilify Maintena® Missed Doses				
Time Since Last Injection	If the patient has missed the 2 nd or 3 rd dose of aripiprazole LAI		If the patient has received at least 4 consecutive doses	
	<5 weeks since the last injection	>5 weeks since the last injection	<6 weeks since the last injection	>6 weeks since last injection
Action	Give the next injection as soon as possible	Give the next injection as soon as possible plus oral aripiprazole for 2 weeks	Give the next injection as soon as possible	Give the next injection as soon as possible plus oral aripiprazole for 2 weeks
Abilify Asimtufii® Missed Doses				
Time Since Last Injection	8-14 weeks since last dose		>14 weeks since last dose	
Action	Administer Asimtufii® as soon as possible, then resume 2-monthly schedule		Administer Asimtufii® as soon as possible, but in addition give oral aripiprazole 10-20mg daily for 14 days. Then resume 2-monthly schedule	

Approximate dose equivalence between oral and long-acting injection aripiprazole formulations

Aripiprazole oral	Abilify Maintena® 1 monthly LAI	Abilify Asimtufii® 2 monthly LAI
10-15 mg/day	300mg/4 weeks	720mg/8 weeks
15-20 mg/day	400mg/4 weeks	960mg/8 weeks

5.7 Typical Antipsychotics

- Typical Antipsychotic LAIs Test doses of the typical antipsychotic LAI are recommended prior to therapy to ensure tolerability of the medication and the oily vehicle.

Medication	Available products	Licensed injection site	Initiation regime	Licensed maintenance dose range	Accepted exceptional dosing window	Storage
Flupentixol (Flupenthixol) decanoate (Dissolved in coconut oil)	Fluanxol® Depot 20mg in 1mL, 40mg in 1 mL Fluanxol® Concentrated Depot 100mg in 1 mL	Gluteal	Test dose of 5mg - 20mg to be given 4-10 days prior to maintenance dosing to ensure tolerability to oily vehicle and medication	20mg – 100mg every 2 – 4 weekly	Nil specified in the product information. 1-2 days before or after the due date is acceptable	Store below 25°C Protect from Light
Haloperidol decanoate (Dissolved in sesame oil)	Haldol® Decanoate 50mg in 1 mL, 150mg in 3mL	Gluteal	Test dose of 25mg to be given 4-10 days prior to maintenance dosing to ensure tolerability to oily vehicle and medication	25mg - 300mg every 4 weekly	Nil specified in the product information. 1-2 days before or after the due date is acceptable	Store below 25°C Protect from Light
Zuclopenthixol decanoate (Dissolved in coconut oil)	Clopixol® Depot 200mg in 1mL	Gluteal	Test dose of 100mg to be given 4-10 days prior to maintenance dosing to ensure tolerability to oily vehicle and medication	200mg – 400mg every 2 - 4 weekly	Nil specified in the product information. 1-2 days before or after the due date is acceptable	Store below 25°C Protect from Light
DO NOT CONFUSE WITH ZUCLOPENTHIXOL ACETATE (CLOPIXOL ACUPHASE®)						

5.8 Preparation and Administration

The atypical antipsychotic depots are pre-dissolved in an oily base ready for drawing up and administering. Available products do not come supplied with needles

- Flupentixol (Flupenthixol)
 - Product information does not specify. Clinician discretion is advised. A 2 inch (50mm) 21 G needle is acceptable
- Haloperidol
 - Use 2 inch (50mm) 21 G needle
- Zuclopenthixol
 - Product information does not specify. Clinician discretion is advised. A 2 inch (50mm) 21 G needle is acceptable.

5.9 Missed Doses

Haloperidol decanoate (Haldol Decanoate®) Missed Doses			
Time since Last Injection	Steady state not reached (less than 2 months after commencement ³) or it has been >6 to 12 weeks since last dose	At steady state (at least 2-4 months after commencement) and ≤6 weeks since last injection	It has been ≥13 weeks since the last injection
Action	• Give the next injection as soon as possible • Provide oral antipsychotic supplementation if symptoms reoccur • Around Day 6 after injection (time to peak), monitor closely for adverse effects	Next injection should be given as soon as possible	• Stabilise on oral haloperidol ○ The dosage equivalent between oral and depot form is 10-15 times. For example, haloperidol decanoate 150mg monthly depot is equivalent to oral haloperidol 10-15mg daily³ • Haloperidol decanoate should be reinstituted

Zuclopenthixol Decanoate (Clopixol Depot®) Missed Doses			
Time since Last Injection	At steady state (at least 14 weeks after commencement ⁴) and ≤4 weeks since last injection	Steady state has not been reached (less than 14 weeks after commencement ²) or it has been >4 to 12 weeks since last dose	It has been ≥12 weeks since the last injection
Action	Next injection should be given as soon as possible	• Give the next injection as soon as possible • Provide oral antipsychotic supplementation if symptoms reoccur • Around Day 7 after injection (time to peak), monitor closely for adverse effects	• Stabilise on oral zuclopenthixol • The dosage equivalent between oral and depot form is 8 times. For example, zuclopenthixol decanoate 200mg fortnightly depot is equivalent to oral zuclopenthixol 25mg daily⁴ • Zuclopenthixol decanoate should be reinstituted

Flupentixol Decanoate (Fluanxol Depot® and Fluanxol Concentrated Depot®) Missed Doses			
Time since Last Injection	Steady state not reached (less than 3 months after commencement ⁵) or it has been >3 to 10 weeks since last dose	At steady state (at least 3-6 months after commencement ⁵) and ≤3 weeks since last injection	It has been ≥10 weeks since the last injection
Action	• Give the next injection as soon as possible • Flupentixol (Flupentixol) is not available in oral form in Australia. If symptoms reoccur, provide oral antipsychotic supplementation with another antipsychotic • Around Day 2 after injection (time to peak), monitor closely for adverse effects	• Next injection should be given as soon as possible	• Stabilise on another oral antipsychotic as flupentixol is not available in Australia in an oral formulation • Flupentixol (Flupentixol) decanoate should be reinstituted



6. Monitoring and Evaluation

This Guideline will be evaluated as required to determine its currency, relevance, effectiveness, efficiency, and sustainability. At a minimum, it will be reviewed every three (3) using a risk management approach in accordance with the [NMHS Policy Governance Framework](#).

7. Definitions

Terms	Definitions
Depot injection or LAI	A sustained-action medication formulation given by deep intramuscular injection, allowing slow release and gradual absorption so that the active agent can act for much longer periods than is possible with standard injections. For the purposes of this document the terms depot and LAI will be used interchangeably.
Z-Track Technique	A type of intramuscular injection technique used to prevent leakage of the medication into the subcutaneous tissue.
Steady State	- Steady state concentration is the time during which the medication's plasma level remains stable when the medication is given repeatedly, i.e. when there is a balance between a medication being administered and the medication being excreted or eliminated. - Steady state is reached after 4 to 5 half-lives of the medication. - Most depots are administered at a frequency of approximately one half-life of the depot formulation.

Refer to [NMHS Policy Document Glossary of Terms](#) for any other frequently used terms

8. References

West J, Marcus S, Wilk J, Countis L, Regier D, Olfson M. Use of Depot Antipsychotic Medications for Medication Nonadherence in Schizophrenia. *Schizophrenia Bulletin* 2008;34(5):995-1001.

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RISPERDAL CONSTA® (Risperidone): Use of High Doses. Communication from Janssen Cilag Pty Ltd. Version: 26 Sept 2011.

MIMS Online for individual medication monographs

Bazire, S. (2020). *Psychotropic Drug Directory 2020/21*. London, UK: Park Communications Limited

Taylor, D., Barnes, T., Young, A. (2021). *The Maudsley Prescribing Guidelines in Psychiatry* (14th ed.). Hoboken, NJ: Wiley

9. Related Documents

NMHS MHS Policy documents	NMHS MH Medication: Prescribing, Administration and Management In The Community NMHS MH Medication: Prescribing, Administration and Management For Inpatients NMHS MH Medication: Off-Label Prescribing Including Prescribing Greater than Maximum Dose Policy
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10. Document Control

Document Version Control			
Date	Version	Author	Notes
5/06/2019	2.0	MHEG	Endorsed out of session
14/06/2019	2.0	Executive Director	Endorsed out of session – via submission form
14/06/2019	2.0	Chief Pharmacist, NMHS MH	Scheduled review: Update to template, hyperlinks, update to availability of drugs. Updated NSQHS Standards
11/08/2022	3.0	Chief Pharmacist, NMHS MH	New template. Scheduled review. Incorporating the missed doses guidelines
09/01/2025	4.0	Medicines Information Pharmacist, NMHS MH	Added new LAIs Risvan® and Asimtufii®

This document can be made available in alternative formats on request for a person with a disability.

Within the Department of Health WA, the term Aboriginal is used in preference to Aboriginal and Torres Strait Islander, in recognition that Aboriginal people are the original inhabitants of Western Australia. No disrespect is intended to our Torres Strait Islander colleagues and community.

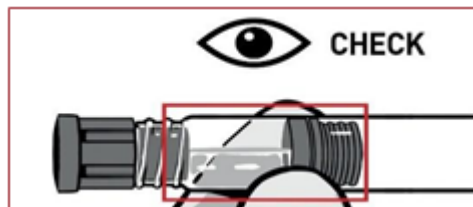
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Appendix 1 – Risperidone Risvan® reconstitution

Each kit box of Risvan® contains a pre-filled syringe containing powder with a WHITE plunger and WHITE finger flange, marked with an “R”. This is the risperidone product. It also contains a pre-filled syringe containing liquid with a transparent plunger and RED finger flange, marked with an “S”. This is the diluent.

Step 1: Inspect diluent and tap powder syringe to dislodge any potential packed powder



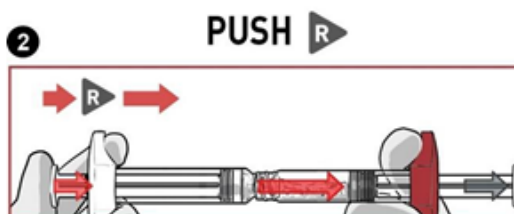
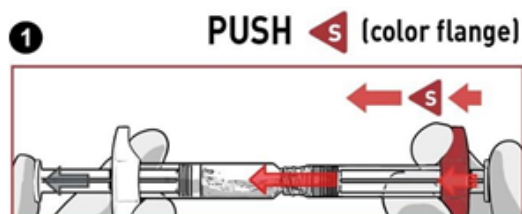
Step 2: Uncap syringes in upright position



Step 3: Connect the syringes, by placing the diluent (red) syringe on top of the powder (white) syringe and twist together.



Step 4: Mix the contents. Push vigorously the diluent (red) syringe towards the powder (white) syringe. Quickly start mixing contents by pushing plungers fast and alternately for 100 pushes (2 pushes/second, approx. 1 minute). Note that the medication is viscous and some force is required when pressing plunger rods.



Step 5: Transfer all contents into the red syringe. Detach syringes, then attach appropriate sterile needle. (21G 1 inch [green cap] for deltoid administration, 20G 2 inch [yellow cap] for gluteal)

Step 6: Inject entire contents of syringe into the deltoid or gluteal muscle. Injection time is longer than usual due to viscosity of the medicine. Wait a few seconds before removing the needle.